

Olist E-Commerce: Order Fulfilment & Customer Satisfaction

Data Analysis | VAC Project Report

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I. INTRODUCTION

The project utilizes the Olist Brazilian E-Commerce Public Dataset, which is a dataset containing information from the biggest Brazilian online department store marketplace that operated from 2016 to 2018. In total, the dataset comprises several files, with each of them having something to do with a particular part of the business: orders, customers, products, sellers, payments, and reviews.

II. OBJECTIVE

The core objective of this project is to perform Exploratory Data Analysis (EDA) on selected datasets to understand what drives customer satisfaction, specifically, whether late deliveries lead to lower review scores, and whether payment method or total payment value has any relationship with how a customer rates their experience.

The analysis follows a structured pipeline:

- Load and understand the relevant datasets
- Merge them into a single table
- Clean the data by removing nulls and outliers
- Feature Engineering
- Perform univariate & bivariate analysis on variables
- Create a correlation matrix
- Summarise key findings

Dataset used – Olist Brazilian Dataset

- olist_orders_dataset.csv – the lifecycle of every order (purchase date, delivery date, estimated date, status)
- olist_order_reviews_dataset.csv – the review score (1–5) left by the customer after delivery
- olist_order_payments_dataset.csv – how the order was paid (method, total value)
- olist_customers_dataset.csv – the state the customer is located in

III. IMPLEMENTATION

1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import warnings
```

2. Load & Merge Relevant Datasets

```
orders = pd.read_csv("/content/olist_orders_dataset.csv",
                    parse_dates=["order_purchase_timestamp",
                                "order_approved_at",
                                "order_delivered_carrier_date",
                                "order_delivered_customer_date",
                                "order_estimated_delivery_date"])
reviews = pd.read_csv("/content/olist_order_reviews_dataset.csv",
                    parse_dates=["review_creation_date",
                                "review_answer_timestamp"])
payments = pd.read_csv("/content/olist_order_payments_dataset.csv")
customers = pd.read_csv("/content/olist_customers_dataset.csv")
```

We use `parse_dates`, which tells pandas to automatically convert date columns into proper datetime objects at the time of loading.

The four date columns parsed are: `order_purchase_timestamp`, `order_delivered_customer_date`, and `order_estimated_delivery_date`.

3. Understanding the Data

```
print(f"orders : {orders.shape}")
print(f"reviews : {reviews.shape}")
print(f"payments : {payments.shape}")
print(f"customers : {customers.shape}")

reviews_clean = (reviews
                .sort_values("review_creation_date")
                .drop_duplicates(subset="order_id", keep="last")
                [["order_id", "review_score"]])

df = pd.merge(df,
              customers[["customer_id", "customer_state"]],
              on="customer_id", how="left")
```

4. Feature Engineering

```
df["delivery_days"] = (  
    (df["order_delivered_customer_date"] -  
     df["order_purchase_timestamp"]).dt.days  
)  
df["delay_days"] = (  
    (df["order_delivered_customer_date"] -  
     df["order_estimated_delivery_date"]).dt.days  
)  
df["is_late"] = (df["delay_days"] > 0).astype(int)
```

delivery_days - Calculated as the number of days between order_purchase_timestamp and order_delivered_customer_date.

delay_days - Calculated as the difference between actual delivery and estimated delivery. A positive value means the order arrived late, a negative value means it arrived early.

is_late - 1 if delay_days is positive (late), 0 if not.

5. Data Cleaning

```
print("\n Missing values before cleaning -")  
print(df.isnull().sum()[df.isnull().sum() > 0])  
  
df.dropna(subset=["delivery_days", "delay_days",  
                 "review_score", "total_payment"], inplace=True)
```

After filtering, some rows may still have NaN in key columns. dropna(subset=[...]) is used to remove only rows where at least one of the core columns is empty.

```
for col in ["delivery_days", "total_payment"]:  
    Q1 = df[col].quantile(0.25)  
    Q3 = df[col].quantile(0.75)  
    IQR = Q3 - Q1  
    df = df[df[col].between(Q1 - 3 * IQR, Q3 + 3 * IQR)]  
  
df["review_score"] = df["review_score"].astype(int)  
df.reset_index(drop=True, inplace=True)
```

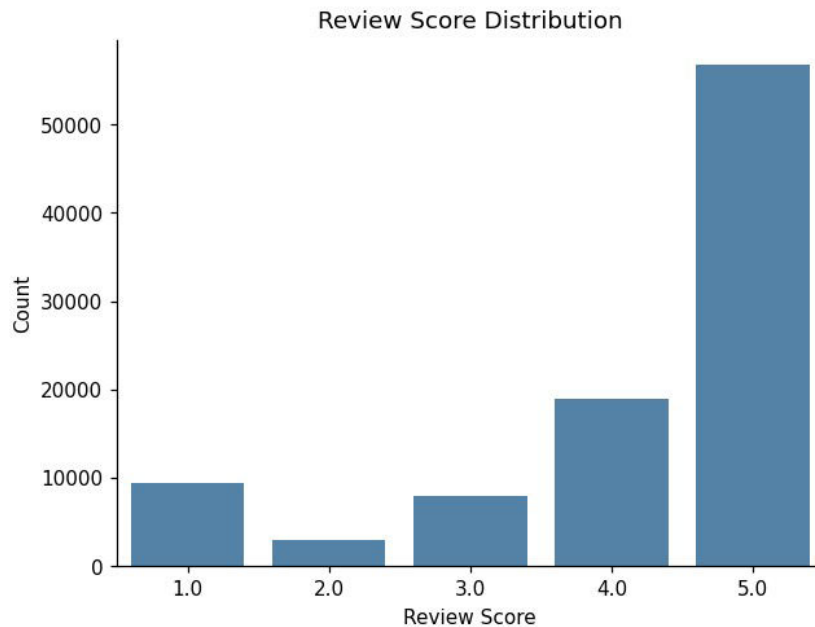
Some orders have extreme values that are likely data errors or genuinely unusual cases. If kept, they pull the mean & distort correlations. The IQR (Interquartile Range) method is a standard way to identify and remove these.

6. Univariate Analysis

Perform univariate analysis on individual variables.

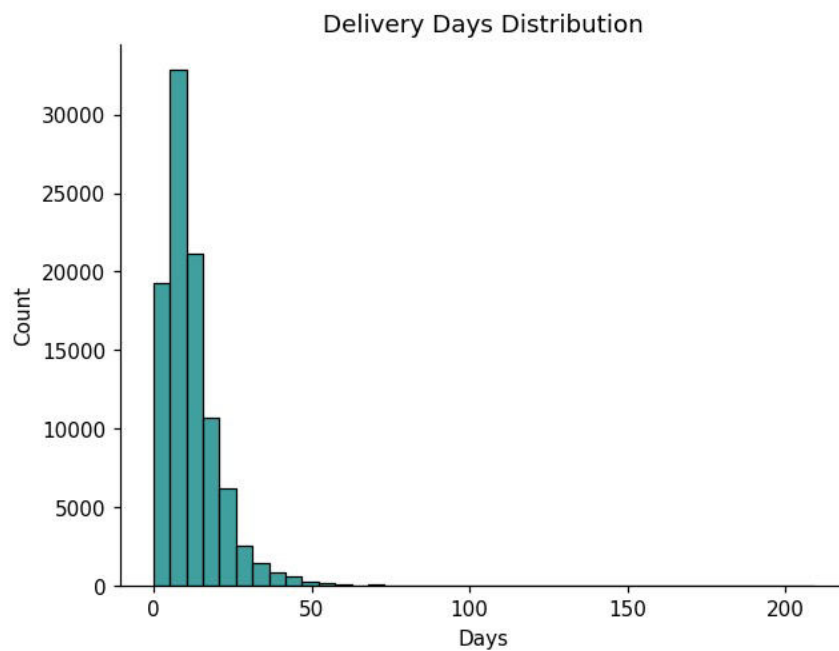
Review Score – Bar Chart

It shows whether score 5 dominates, or whether low scores are common.



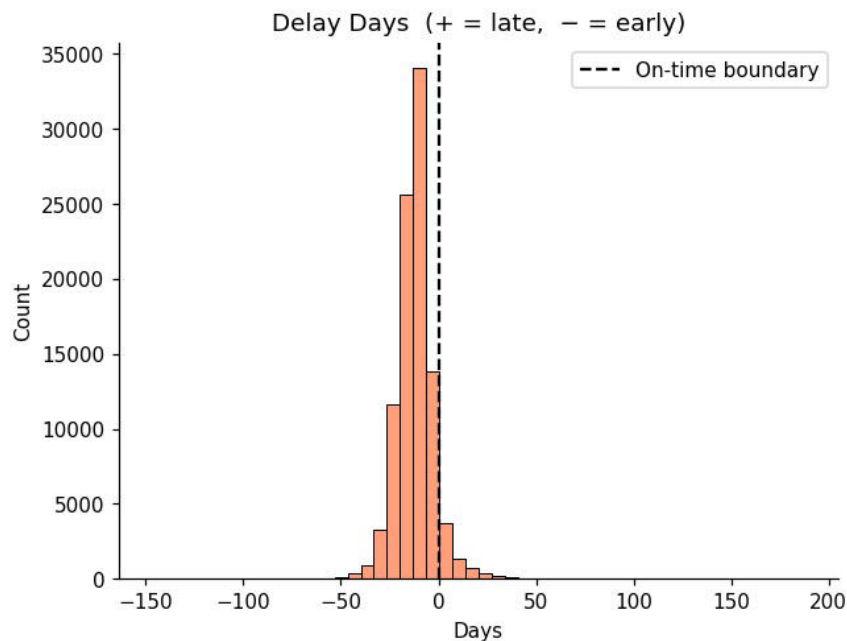
Delivery Days – Histogram

What the typical delivery time is and whether the distribution is symmetric or skewed to one side.



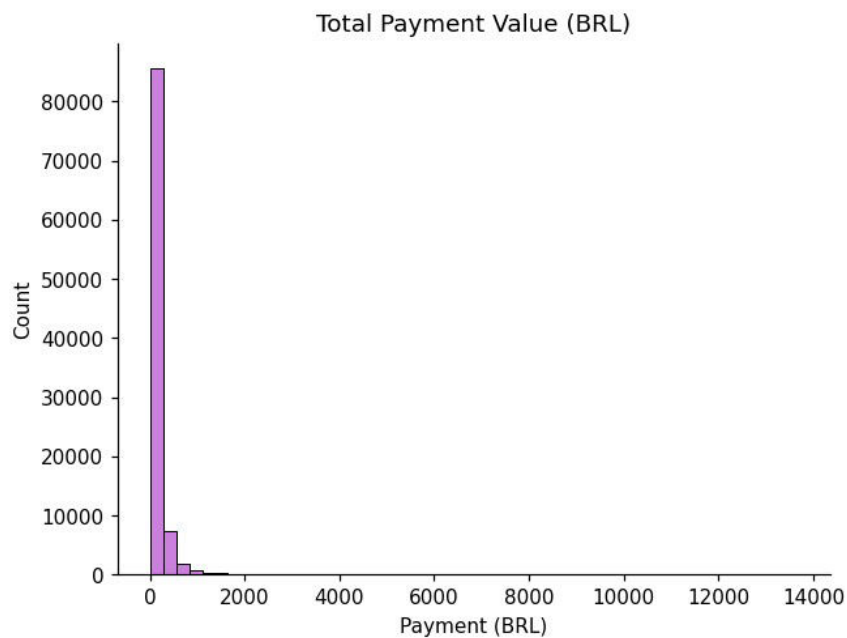
Delay Days – Histogram

Similar to delivery days, but the x-axis can be negative (early delivery) or positive (late).



Total Payment – Histogram

Payment values are expected to be right-skewed — most orders are small, but some are large.

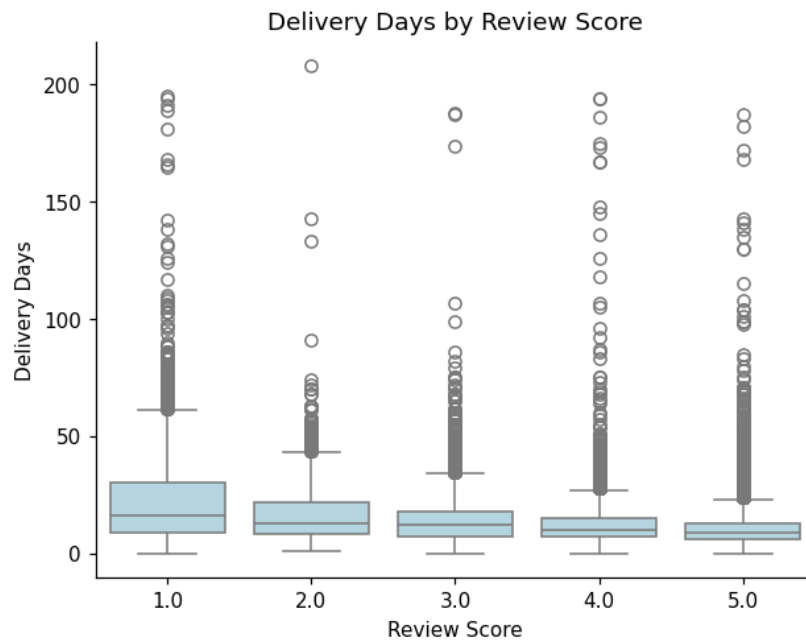


7. Bivariate Analysis

Here, everything is compared against review_score (the target variable).

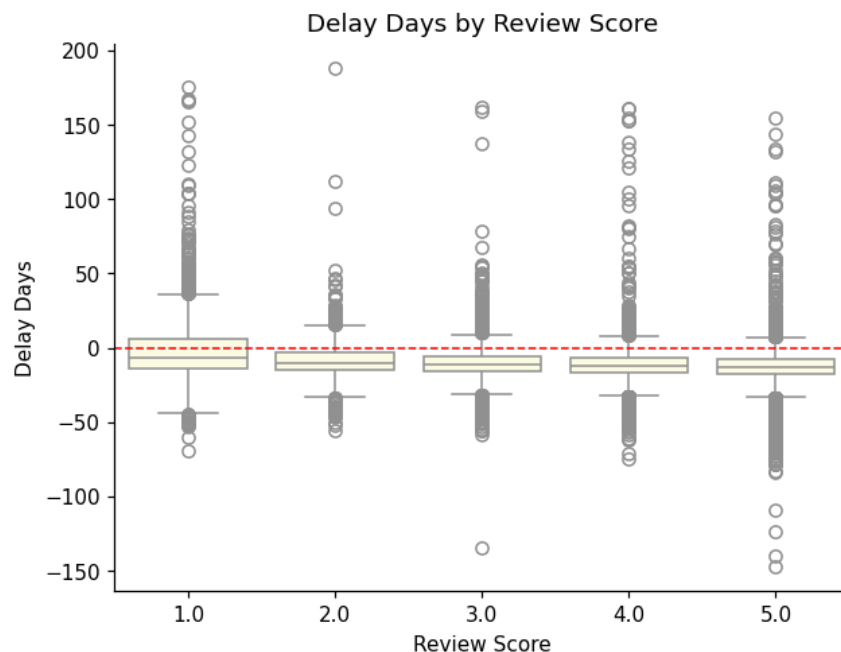
Delivery Days by Review Score – Box Plot

We can directly see whether orders with higher scores had shorter delivery times. If the boxes shift downward as the score increases, that confirms the relationship.



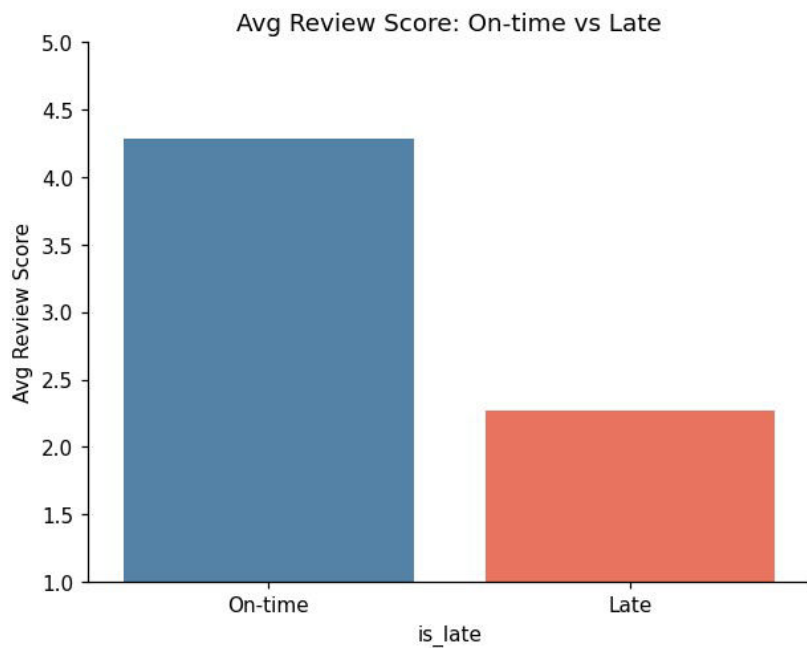
Delay Days by Review Score – Box Plot

It shows whether the customer was disappointed (order arrived later than promised).

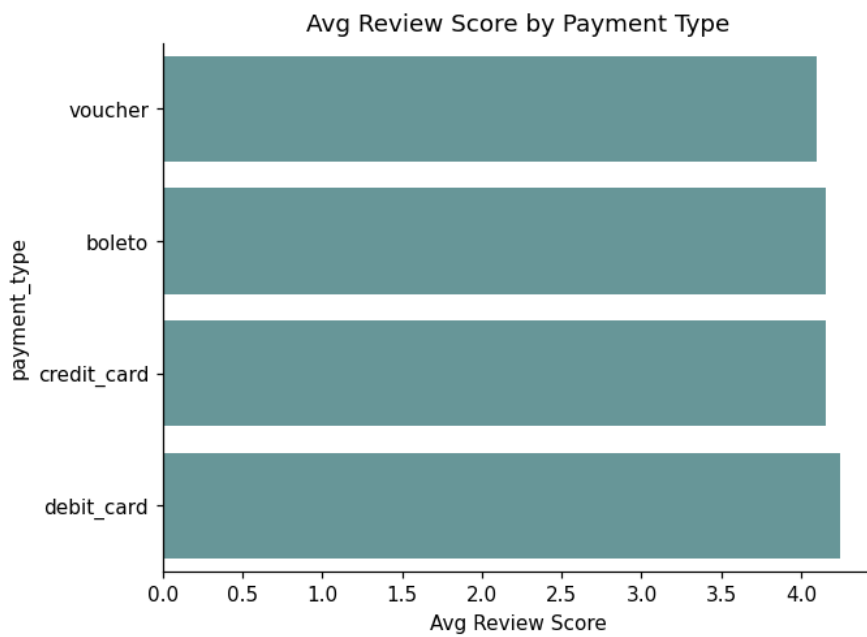


On-time vs Late – Bar Chart

Most direct visual test of whether lateness hurts satisfaction.



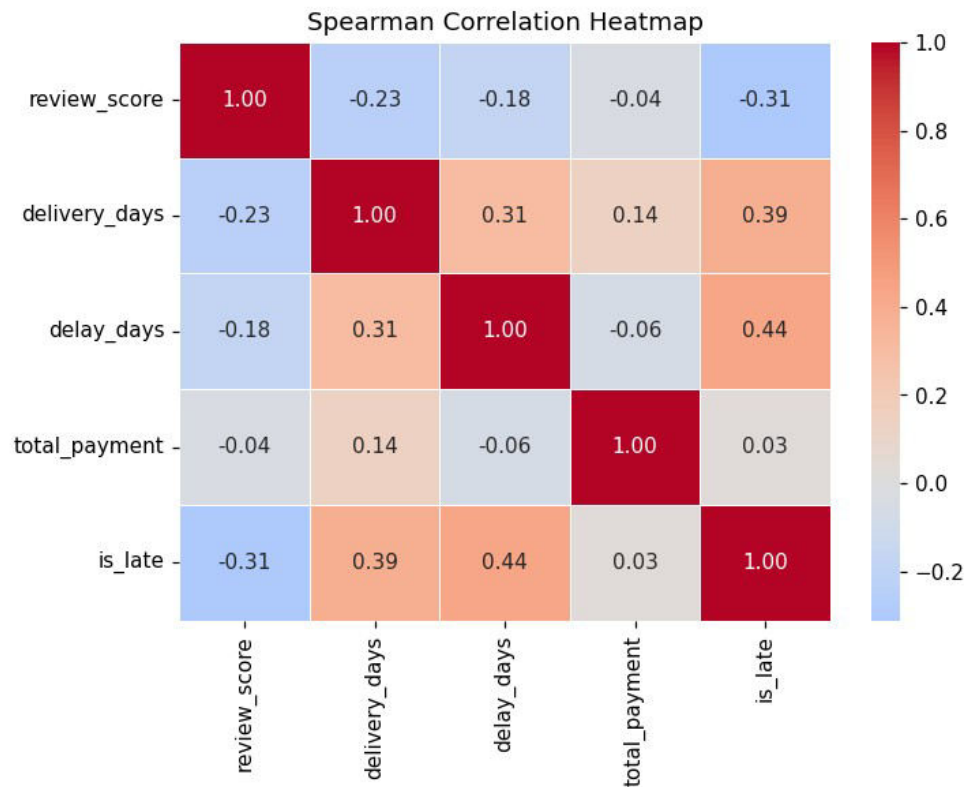
Payment Type vs Average Review Score — Horizontal Bar Chart



8. Correlation Analysis & Heatmap

Correlation is a number between -1 and $+1$ that measures how strongly two variables move together.

The correlation matrix is visualised as a heatmap using seaborn's `sns.heatmap()`. A heatmap colours each cell according to the correlation value.



IV. RESULT

The analysis provides the following insights :-

- **Delivery and satisfaction are linked** - Reviews for orders that came late were much lower on average than reviews for orders that came on time. The box plots for both `delivery_days` and `delay_days` showed that scores went down as delivery time or delay time went up. This shows that delivery speed is one of the most important factors in this dataset that makes customers happy.
- **Quite a few orders arrived late** - The `delay_days` histogram showed that a large number of delivered orders had positive delay values, which means they arrived after the estimated date. This is a problem with operations: either the estimated delivery dates are too optimistic, or the logistics aren't working as well as they should.
- **Delivery times are right-skewed** - Most orders were delivered in a reasonable amount of time, but there were some that took a long time to arrive, which raised the average. The histogram made this skew very clear.
- **Type of Payment has little effect on satisfaction** - The average review score for all payment types was pretty close to the same, with no one payment type

being much higher or lower than the overall mean. This means that how a customer pays doesn't have much of an effect on how happy they are.

- **Correlation Heatmap shows delay as most important variable** - The Spearman heatmap showed that the review score was most strongly (negatively) related to both `delay_days` and `is_late`. This supports the idea that what matters most to customers is the difference between what was promised and what actually happened, not the delivery time itself.

V. CONCLUSION

This project showed a full EDA pipeline, from raw CSV files to useful insights.

The analysis showed that being late with deliveries is the biggest sign of a low review score. Customers who got their orders later than the expected date always gave lower ratings.

Making delivery time estimates more accurate or improving logistics to meet current estimates would probably directly improve customer satisfaction scores.

On the other hand, payment behaviour had little to do with satisfaction, which means that the checkout and payment process isn't a big problem for customers.